

JEE MAIN REVISION MASTER: TRIGONOMETRIC EQUATIONS OUT OF SYLLABUS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 59 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes	33 PYQs	Quantitative Analysis
Very High	Trigonometric Functions And Equations — Applied Cases, Reactions & Traps	15 PYQs	Applied Topics
High	Trigonometric Functions And Equations — Core Principles & Governing Laws	11 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes	$S=\{\theta \in [-4\pi,4\pi]:3\cos^2\theta+6\cos\theta-10\cos^2\theta+5=0\}$, $\quad n(S)=5$, $\quad (-\frac{\pi}{6})$
Trigonometric Functions And Equations — Applied Cases, Reactions & Traps	$2\sin^2\theta-\cos2\theta=0$, $\quad \frac{3}{4}$, $\quad 2\theta-\cos^2\theta+\sqrt{2}=0$
Trigonometric Functions And Equations — Core Principles & Governing Laws	$14\operatorname{cosec}^2x-2\sin^2x=21-4\cos^2x$, $\quad \frac{\pi}{2}$, $\quad S=\{\theta \in [0,2\pi):\tan(\pi\cos\theta)+\tan(\pi\sin\theta)=0\}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Trigonometric Functions And Equations — Standard Calculations & Problem Archetypes (Very High Yield — 33 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $S = \theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta$. Then $\underset{\theta \in S}{\sum} \theta$ is equal to :..." [Ans: N/A]
- [2026 Morning Shift] Q#3 (MCQ): "Let $S = x \in [-\pi, \pi] : \sin x (\sin x + \cos x) = a, a \in \blacksquare$. Then $n(S)$ is equal to :..." [Ans: D]
- [2026 Morning Shift] Q#4 (MCQ): "Number of solutions of $\sqrt{3}\cos 2\theta + 8\cos\theta + 3\sqrt{3} = 0, \theta \in [-3\pi, 2\pi]$ is :..." [Ans: N/A]

3.2 Trigonometric Functions And Equations — Applied Cases, Reactions & Traps (Very High Yield — 15 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Morning Shift] Q#2 (MCQ): "The sum of all the integral values of p such that the equation $3\sin^2 x + 12\cos x - 3 = p, x \in \blacksquare$, has at least one solution, is :..." [Ans: N/A]
- [2025 Evening Shift] Q#2 (MCQ): "The sum of all values of $\theta \in [0, 2\pi]$ satisfying $2\sin^2 \theta = \cos 2\theta$ and $2\cos^2 \theta = 3\sin \theta$ is..." [Ans: N/A]
- [2024 Evening Shift] Q#2 (MCQ): "The sum of the solutions $x \in \blacksquare$ of the equation $\frac{3\cos 2x + \cos^3 2x \cos^6 x - \sin^6 x}{x^3 - x^2 + 6}$ is..." [Ans: N/A]

3.3 Trigonometric Functions And Equations — Core Principles & Governing Laws (High Yield — 11 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Trigonometric Functions And Equations.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Trigonometric Functions And Equations.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Trigonometric Functions And Equations PYQs.

Representative Source PYQs:

- [2025 Morning Shift] Q#3 (MCQ): "The number of solutions of the equation $2x + 3\tan x = \pi, x \in [-2\pi, 2\pi] - \{\pm(\pi/2), \pm(3\pi/2)\}$ is..." [Ans: N/A]
- [2024 Morning Shift] Q#3 (MCQ): "Let $|\cos\theta \cos(60-\theta)\cos(60+\theta)| \leq (1/8), \theta \in [0, 2\pi]$. Then, the sum of all $\theta \in [0, 2\pi]$, where $\dots$ " [Ans: N/A]
- [2023 Evening Shift] Q#3 (Numerical): "Let $S = \{\theta \in [0, 2\pi) : \tan(\pi \cos\theta) + \tan(\pi \sin\theta) = 0\}$. Then $\text{card}\{ \theta \in S : \sin^2(\theta + (\pi/4)) \text{ is } \dots \}$ " [Ans: 2]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$. Then $\text{card}\{ \theta \in S : \sin\theta \}$ is equal to :

- (A) $-(2\pi/3)$
- (B) $-(4\pi/3)$
- (C) $(2\pi/3)$
- (D) $(4\pi/3)$

Correct Answer: None

Solution Analysis: Given, $\cos\theta + 1 = \sqrt{3}\sin\theta$ We have to find all $\theta \in (-2\pi, 2\pi)$ satisfying this equation, and then calculate their sum. Rewrite the equation as $1 + \cos\theta = \sqrt{3}\sin\theta$ Now use the standard identities: $1 + \cos\theta = 2\cos^2(\theta/2)$ and $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$ Substituting these, $2\cos^2(\theta/2) = \sqrt{3} \cdot 2\sin(\theta/2)\cos(\theta/2)$ So, $2\cos(\theta/2)(\cos(\theta/2) - \sqrt{3}\sin(\theta/2)) = 0$ Hence, either Case 1: $\cos(\theta/2) = 0$ So, $(\theta/2) = (\pi/2) + n\pi$ which g

Benchmark Problem 2: JEE Main 2026 (Online) 5th April Morning Shift (Q#2)

The sum of all the integral values of p such that the equation $3\sin^2x + 12\cos x - 3 = p, x \in \mathbb{R}$, has at least one solution, is:

- (A) -54
- (B) -60
- (C) -75
- (D) -84

Correct Answer: None

Solution Analysis: Given $3\sin^2x + 12\cos x - 3 = p, x \in \mathbb{R}$ We need all integral values of p for which this equation has at least one real solution. Now use the identity $\sin^2x = 1 - \cos^2x$ So, $3(1 - \cos^2x) + 12\cos x - 3 = p$ Simplifying, $3 - 3\cos^2x + 12\cos x - 3 = p$ $-3\cos^2x + 12\cos x = p$ Let $t = \cos x$ Then $-1 \leq t \leq 1$ So we have $p = -3t^2 + 12t$ Now we need the range of $f(t) = -3t^2 + 12t, -1 \leq t \leq 1$ This is a downward opening parabola. Find its values at the endpoints: $f(-1) = -3(1) + 12(-1) = -3 - 12 = -15$ $f(1) = -3(1) + 12(1) = -3 + 12 = 9$ Also, $f'(t) = -6t + 12$ For $-1 \leq t \leq 1$, we have $-6t + 12 > 0$ so $f(t)$ is increasin

Benchmark Problem 3: JEE Main 2025 (Online) 3rd April Morning Shift (Q#3)

The number of solutions of the equation $2x + 3\tan x = \pi$, $x \in [-2\pi, 2\pi] - \{\pm(\pi/2), \pm(3\pi/2)\}$ is:

- (A) 4
- (B) 5
- (C) 3
- (D) 6

Correct Answer: None

Solution Analysis: $2x + 3\tan x = \pi \Rightarrow \tan x = (\pi - 2x/3)$, $x \in [-2\pi, 2\pi] - \{\pm(\pi/2), \pm(3\pi/2)\}$ 5 solutions